

LLM Safety: Jailbreaks (Instruction Bypass), Prompt Injection, and Hallucination Robustness

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1. In-Depth Review

The answers to the following questions are mandatory and will be shared with both the coordinator and the authors.

What is this paper about, what contributions does it make, and what are the main strengths and weaknesses?

Please describe what problem or question this paper addresses and the main contributions it makes towards a solution or answer. (Further, you will include this paper's main strengths and weaknesses.)

The paper addresses safety evaluation of LLMs. It focuses on three critical robustness dimensions: jailbreaking, prompt injection, and hallucination robustness. The authors evaluate multiple open-source models such as LLaMA 3.1 8B, Mistral 7B, Qwen 3 8B and one proprietary model GPT-5-Nano using curated benchmarks and custom prompt sets.

The main contributions are:

1. A unified evaluation pipeline covering three distinct but interrelated LLM safety risks.
2. Empirical evidence of safety-utility trade-offs across models, highlighting how stricter alignment often reduces benign task performance.
3. A reproducible experimental setup using local inference, fixed seeds, and standardized datasets.

Strong points (at least two)

What are the main strong points - contributions of the work?

1. Instead of looking at just one issue, the project studies three major safety risks: jailbreaking, prompt injection, and hallucinations. This gives a broader and more realistic picture of how language models behave in unsafe situations.

2. The results make it easy to see how different models behave: LLaMA is very cautious but often refuses harmless requests, Mistral is helpful but less safe, and Qwen sits somewhere in between. The use of confusion matrices and category-level breakdowns helps explain these differences clearly.
3. The experiments are run locally, use fixed settings, and rely on well-known public benchmarks. This makes it easier for others to understand what was done and to repeat or extend the experiments.

Weak points (at least two)

What are the weakest points of the work?

1. The experiments use relatively small sets of prompts (usually between 40 and 100 per category). The results are mostly presented as counts or percentages, without deeper statistical analysis, so it's hard to judge how stable or general the findings are.
2. Many decisions, such as whether a response counts as a refusal or a successful attack, are made manually or semi-manually. Since there are no clear labeling rules or agreement checks between different reviewers, some bias may be introduced.

Additional suggestions and questions for the Authors to improve their work and their 1ST PROJECT FINAL REPORT (e.g. you can add more ablation tests on ...).

N/A

Missing references, ethical concerns

Some citations are incomplete or contain placeholders like "2025; ?; ?", which should be resolved.

The paper uses open-source models for harmful prompt testing in a responsible manner, which is good. However, using an LLM to grade hallucinations introduces evaluation bias, which should be acknowledged more explicitly.

Work limitations comments

Hallucination evaluation depends on automated grading, which may itself hallucinate or misjudge subtle errors.

Reproducibility criteria

Has clear description of datasets, prompt counts, models, and environment.
Local inference and fixed seeds improve reproducibility.

Clean code comments

The code is structurally clean and readable, but could use more docstrings to make it more comprehensive for others as the codebase is quite large.

Overall Recommendation: 4.5

Reviewer Confidence: 4

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